# Project Documentation

Project Name: **Kind Meal – Turning Excess into Access**
Team Members: **Anubhav Ghosh (23 BCA 10303)**

---

## Project Description

The **Kind Meal Android Application** is designed to address the growing problem of **food wastage** by creating a digital bridge between **restaurants with surplus food** and **NGOs or Trusts serving underprivileged communities**.

Every day, restaurants, hotels, and food outlets generate large quantities of edible food that often goes to waste. At the same time, NGOs and charitable organizations face difficulties in meeting the nutritional needs of the communities they support. **Kind Meal provides a unified solution** to connect these two sectors efficiently.

The application has been developed in **Android Studio using Kotlin and XML**, along with **Jetpack Compose** for modern UI components. A **custom library mix** has been used to improve UI/UX flexibility. For the backend, **Firebase** enables **user authentication, real-time database management, and cloud notifications**, ensuring smooth and secure operations between donors and NGOs.

In the future, Kind Meal aims to expand beyond restaurants and hotels to include **households, events, supermarkets, and catering services**, thereby addressing food wastage across the entire food sector.

---

## Project Objectives

- To develop a user-friendly Android app in **Kotlin (XML + Jetpack Compose)** for restaurants and NGOs.
- To minimize food wastage by enabling real-time food donations through **Firebase Realtime Database**.
- To ensure secure access with **Firebase Authentication** for both donors and NGOs.
- To provide **status updates**.
- To establish a **scalable platform** that can extend beyond restaurants to cover the entire food sector.

---

## System Workflow & Functionality

The **Kind Meal application** provides a structured and transparent workflow to manage food donations:

1. **Food Donation by Restaurants/Food Sector**
   - Restaurants and food outlets create a **donation entry**.
   - Donation includes:

    - **Food description** (type, quantity, and details).
    - **Photo of the food item(s)**.
    - **Pickup availability details**.

2. **NGO/Trust Access**
    - NGOs/Trusts view **all available donations** in real time.
    - They can review details and food photos before responding.
3. **Donation Request Handling**
    - NGOs can **accept or reject** donations.
    - Upon acceptance, NGOs select a **required delivery date**.
4. **Restaurant Confirmation**
    - Restaurants receive the NGO's request and **review the delivery date**.
    - If approved, the donation is confirmed.
5. **Access and Completion**
    - The donation entry is finalized and **locked for both parties**.
    - NGOs get access for pickup/delivery.
    - **Status updates** allow both sides to track the donation until completion.

This workflow ensures:

- **Transparency** between restaurants and NGOs.
- **Real-time updates** on donation availability.
- **Mutual confirmation** before finalizing donations.
- **Efficient redistribution** of surplus food.

---

## Project Specific Outcomes (PrSO)

**PSO1 – Mobile App Development & UI/UX Design**
Students will gain expertise in **Android Studio, Kotlin, XML, and Jetpack Compose**, learning to design hybrid mobile UIs that combine modern layouts with custom library elements.

**PSO2 – Backend Integration with Firebase**
Students will implement **Firebase Authentication, Realtime Database, and Cloud Messaging** for secure login, real-time food listings, and notifications, enhancing backend skills.

**PSO3 – Real-World Social Impact Application**
The project shows how **technology can address food wastage** by connecting donors and NGOs. Students will learn to create scalable, socially impactful applications that align with sustainability goals.

## CO–PO–PSO–PrSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PSO1 | PSO2 | PSO3 | PrSO1 | PrSO2 | PrSO3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 3 | 2 | 3 | 3 | 3 | 2 | 2 | 1 | 3 |
| CO2 | 2 | 3 | 2 | 3 | 2 | 2 | 2 | 3 | 2 | 1 | 2 |
| CO3 | 2 | 2 | 2 | 3 | 3 | 2 | 3 | 2 | 2 | 1 | 2 |
| CO4 | 2 | 2 | 3 | 2 | 3 | 2 | 2 | 2 | 3 | 1 | 2 |
| CO5 | 1 | 1 | 2 | 1 | 1 | 1 | 2 | 1 | 2 | 2 | 1 |

Legend:

- **1 -** Low contribution
- **2 -** Medium contribution
- **3 -** High contribution
- **CO** - *Course Outcomes*
- **PO** - *Program Outcomes*
- **PSO** - *Program Specific Outcomes*
- **PrSO** - *Project Specific Outcomes*

## Project Gap Course

Modern Android App Development with **Kotlin, Jetpack Compose & Firebase**

**Purpose of the Gap Course**

To provide students with essential knowledge and hands-on skills to build **modern Android applications** using **Kotlin, XML, and Jetpack Compose**, integrated with **Firebase backend services** for real-time data management, authentication, and notifications.
**Link:** https://www.coursera.org/learn/chandigarh-university-36610751-2249-4d3a-8bc8-a5df22a2b19b

---

## Skills Required

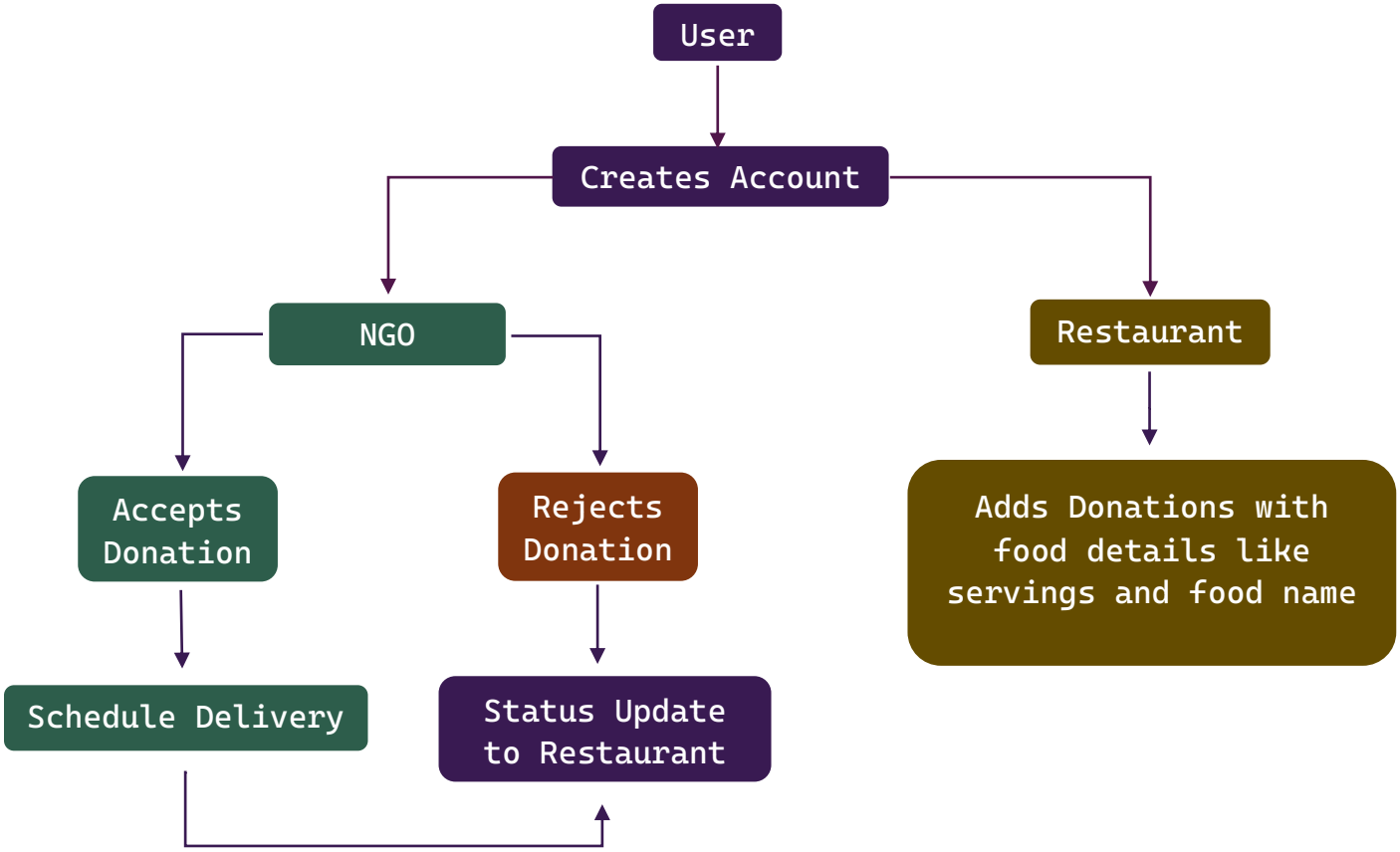
This project requires:

- Kotlin Programming in Android Studio
- **XML and Jetpack Compose** for UI Design

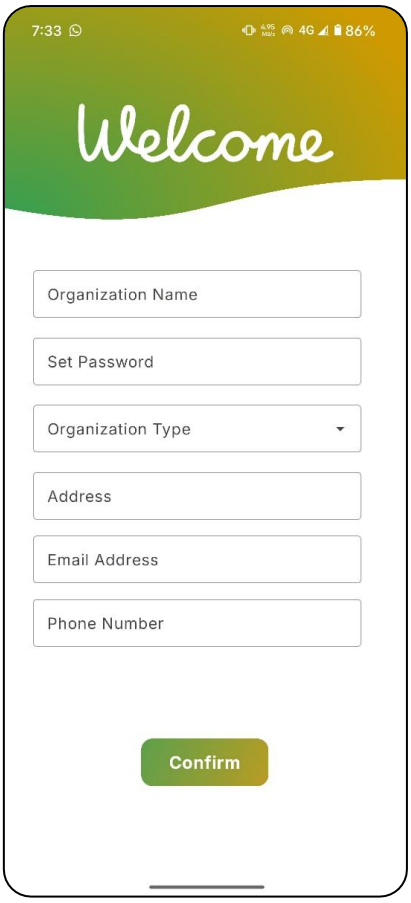
- **Firebase Realtime Database & Authentication**
- **Firebase Cloud Messaging for Notifications**
- **Custom Android Libraries & Dependency Management**
- **UI/UX Design** for Mobile Applications

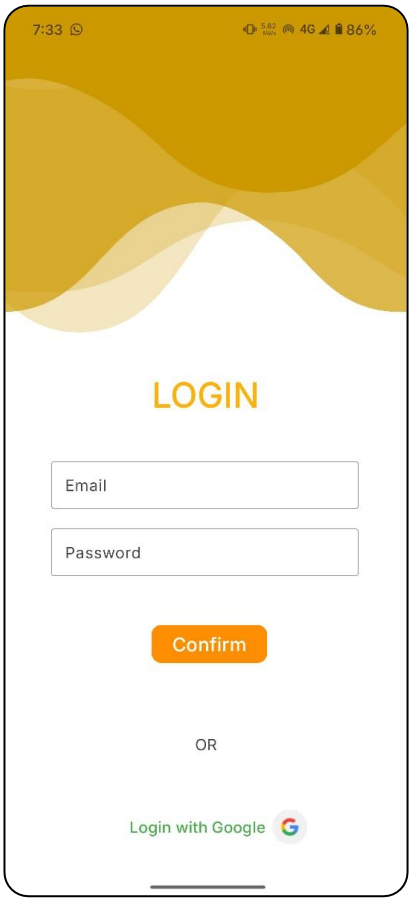
  Problem-solving for Social Applications

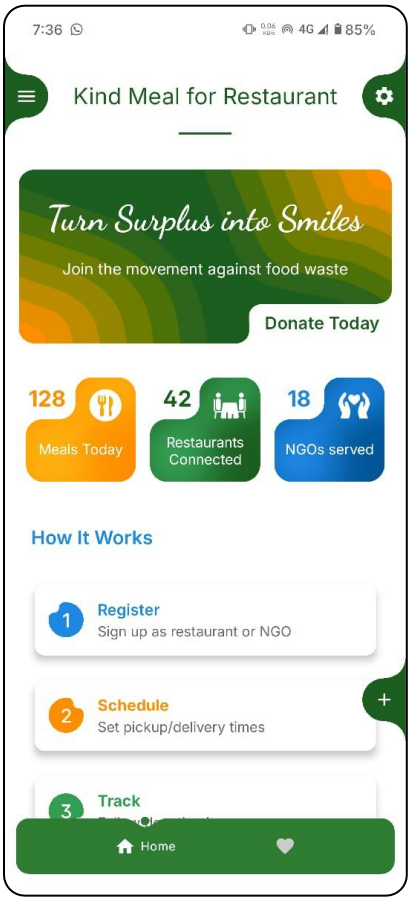
## Project Workflow

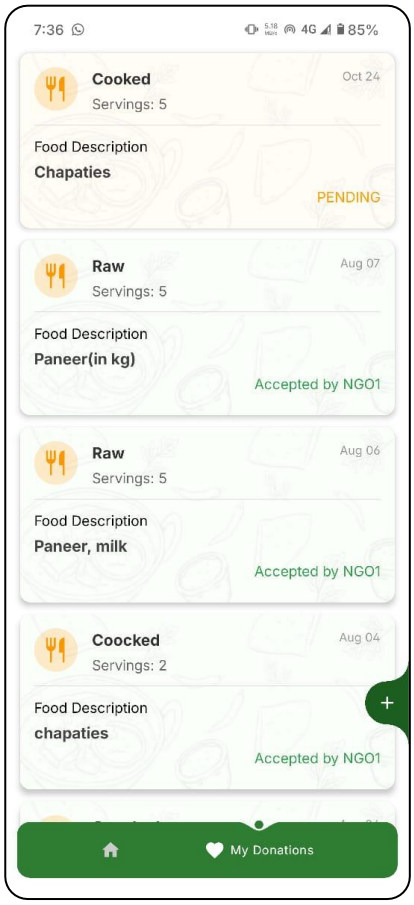


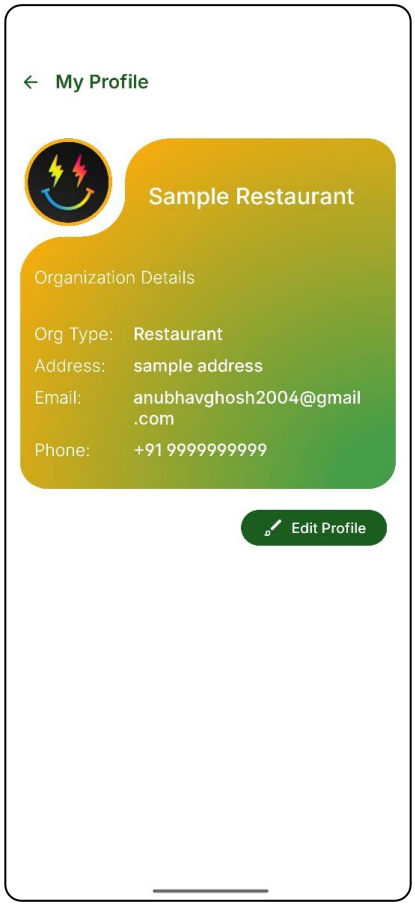
# Application Prototype

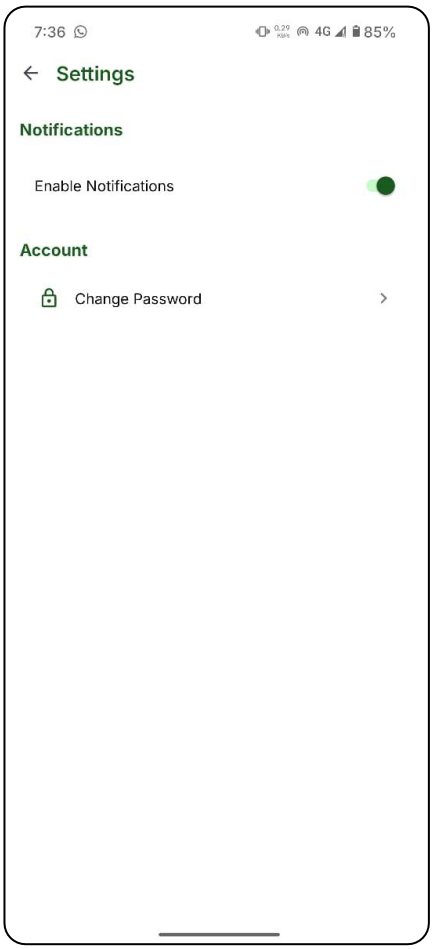


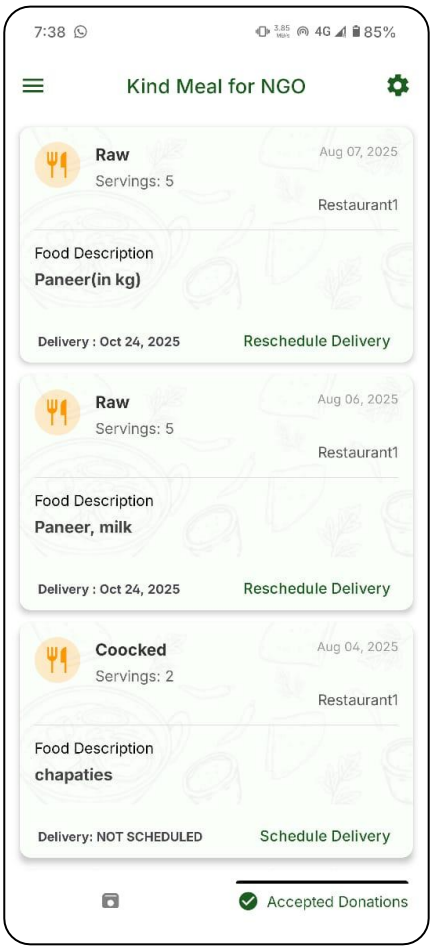



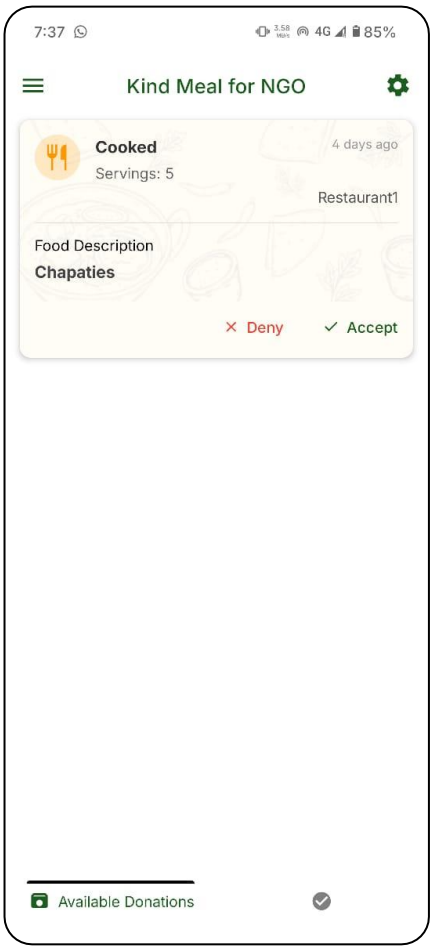




7:36
4G 85%
Settings
Notifications
Enable Notifications
Account
Change Password


7:38
4G 85%
Kind Meal for NGO
Raw
Aug 07, 2025
Servings: 5
Restaurant1
Food Description
Paneer(in kg)
Delivery : Oct 24, 2025
Reschedule Delivery
Raw
Aug 06, 2025
Servings: 5
Restaurant1
Food Description
Paneer, milk
Delivery : Oct 24, 2025
Reschedule Delivery
Coocked
Aug 04, 2025
Servings: 2
Restaurant1
Food Description
chapaties
Delivery: NOT SCHEDULED
Schedule Delivery
Accepted Donations


7:37
4G 85%
Kind Meal for NGO
Cooked
4 days ago
Servings: 5
Restaurant1
Food Description
Chapaties
Deny
Accept
Available Donations